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Assignment-11

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1. **List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.**

Wi-Fi Standard	Frequency Band	Maximum Speed
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Up to 600 Mbps
802.11ac (Wi-Fi 5)	5 GHz	Up to 6.9 Gbps
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz	Up to 9.6 Gbps

Conclusion: Wi-Fi 6 supports higher speeds and better network efficiency than Wi-Fi 4 and Wi-Fi 5.

2. **Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.**



3. **Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.Hint: Consider both signal strength and security when choosing.**

Wi-Fi Network Scan

No.	SSID (Network Name)	Signal Strength	Security Type
1	Home_WiFi	Excellent (90%)	WPA2
2	Office_WiFi	Good (75%)	WPA3
3	Public_WiFi	Weak (40%)	Open (No Password)

Which Network Would I Choose?

I would choose Office_WiFi because it has good signal strength and WPA3 security, which provides stronger protection than WPA2. I would avoid Public_WiFi because it has a weak signal and no password protection.

Note: These are sample observations. Scan nearby Wi-Fi networks on your own device and replace these values with your actual results.

4. **Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.**

SSID Hiding: SSID hiding means hiding the Wi-Fi network name so it does not normally appear in the list of available Wi-Fi networks.

- **Advantage: It makes the Wi-Fi network less visible to ordinary users.**
- **Disadvantage: Users must manually enter the Wi-Fi name to connect, and hiding the SSID does not provide strong security.**

Conclusion: Hiding the SSID alone cannot protect a Wi-Fi network. Using WPA2 or WPA3 security with a strong password is better.